

CONCEPCION, Chile

WMO: 856820

Lat: 36.7783S

Lon: 73.0625W

Elev: 17

StdP: 101.12

Time Zone: -4.00 (CHL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.2	2.3	-2.1	3.2	6.9	-0.2	3.7	5.1	14.2	11.7	12.7	12.0	1.7	150	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.9	24.8	17.4	23.6	16.9	22.7	16.4	18.4	22.9	17.7	22.0	17.0	21.2	6.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.9	12.0	19.6	16.0	11.4	19.0	15.2	10.8	18.3	52.0	23.1	49.7	22.2	47.8	21.2	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.2	9.5	8.4	-0.7	28.6	1.0	1.7	-1.4	29.9	-2.0	30.9	-2.6	31.9	-3.3	33.2		
			-1.3	19.8	1.0	0.8	-2.0	20.4	-2.6	20.9	-3.2	21.3	-3.9	21.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	17.0	16.7	15.4	13.1	11.6	10.1	9.3	9.9	10.8	12.2	14.0	15.8
	DBStd	3.24	1.56	1.57	1.74	1.98	2.21	2.26	2.22	2.05	1.91	1.65	1.64	1.66
	HDD10.0	120	0	0	0	2	10	27	40	27	12	2	1	0
	HDD18.3	1964	47	51	91	157	209	247	279	261	225	189	129	79
	CDD10.0	1209	218	186	168	95	59	29	19	24	37	71	121	181
	CDD18.3	12	6	3	1	0	0	0	0	0	0	0	0	2
	CDH23.3	128	55	31	18	2	0	0	0	0	2	1	4	16
	CDH26.7	9	4	1	2	0	0	0	0	0	0	0	1	1
Wind	WSAvg	3.6	4.3	3.8	3.4	3.0	3.0	3.4	3.4	3.6	3.5	3.6	4.1	4.2
Precipitation	PrecAvg	1080	16	17	24	64	166	222	199	155	86	62	35	25
	PrecMax	1532	84	96	127	241	501	561	496	456	257	182	208	135
	PrecMin	591	0	0	0	0	26	3	24	22	7	0	0	0
	PrecStd	269	21	20	25	50	112	107	102	79	49	47	38	31
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.5	25.8	25.2	22.8	19.5	16.8	16.1	18.1	20.2	21.1	23.1	25.0
		MCWB	18.1	17.8	17.2	15.6	14.9	13.2	11.8	12.3	12.9	15.2	16.2	17.4
	2%	DB	24.8	24.1	23.1	20.2	17.3	15.2	14.2	15.2	17.7	19.1	21.2	23.2
		MCWB	17.6	17.2	16.4	14.8	13.9	12.5	11.7	11.5	12.5	14.0	15.5	16.8
	5%	DB	23.5	23.0	21.9	19.0	16.2	14.2	13.8	14.1	16.1	17.9	20.1	22.1
		MCWB	16.9	16.6	15.9	14.3	13.4	12.3	11.5	11.2	12.0	13.5	14.8	16.2
	10%	DB	22.3	22.0	20.8	17.8	15.2	13.8	13.0	13.2	15.0	16.8	19.0	21.1
		MCWB	16.2	16.0	15.5	13.9	13.1	12.1	11.2	11.0	11.5	12.9	14.1	15.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.5	19.4	18.6	17.4	16.2	14.6	13.6	13.8	14.7	16.0	17.2	18.6
		MCDB	24.9	23.3	22.8	20.1	18.0	15.8	14.6	15.8	18.3	20.2	21.6	23.5
	2%	WB	18.4	18.2	17.5	16.2	15.1	13.6	12.9	13.0	13.6	14.8	16.0	17.5
		MCDB	23.2	22.4	21.5	18.5	16.4	14.6	13.9	14.3	16.0	18.2	19.8	22.1
	5%	WB	17.6	17.4	16.7	15.4	14.4	13.0	12.3	12.4	12.8	14.0	15.3	16.7
		MCDB	22.2	21.6	20.3	17.7	15.6	13.9	13.3	13.7	15.0	16.8	19.0	21.0
	10%	WB	16.9	16.6	15.9	14.6	13.7	12.5	11.7	11.7	12.2	13.2	14.7	16.0
		MCDB	21.2	20.7	19.4	16.9	14.9	13.5	12.7	13.0	14.3	15.9	18.2	20.1
Mean Daily Temperature Range	5% DB	MDBR	10.9	10.6	10.6	9.5	7.7	7.1	7.3	7.6	8.7	9.0	9.9	10.5
		MCDBR	13.0	12.8	12.8	11.8	8.9	7.3	7.6	8.5	11.1	11.3	12.1	12.4
		MCWBR	6.8	6.8	7.1	7.1	6.0	5.3	5.3	5.8	7.2	7.2	7.2	6.9
	5% WB	MCDBR	11.6	11.0	11.0	9.6	6.8	5.1	5.4	6.4	9.1	9.9	10.6	11.1
		MCWBR	6.4	6.2	6.6	6.4	5.0	4.2	4.4	4.9	6.3	6.5	6.5	6.4
Clear-Sky Solar Irradiance	taub	0.323	0.326	0.317	0.322	0.329	0.324	0.316	0.324	0.328	0.323	0.317	0.326	
	taud	2.576	2.566	2.596	2.560	2.538	2.573	2.582	2.552	2.509	2.539	2.579	2.551	
	Ebn at Noon	1012	986	957	890	819	795	827	874	929	978	1012	1013	
	Edh at Noon	105	101	91	83	75	67	70	82	98	103	104	108	
All-Sky Solar Radiation	RadAvg	8.27	7.16	5.68	3.79	2.33	1.82	2.11	2.90	4.46	5.88	7.51	8.27	
	RadStd	0.37	0.37	0.27	0.26	0.21	0.24	0.22	0.25	0.30	0.39	0.33	0.40	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+3
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page